EvenUCan

From Idea to 3D-Printed Object

tronic enclosures with AI assistance.
t the AI design the parts list and
perfect enclosure — no CAD

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github.com/bullbrand/EvenUCan

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CHAPTER 1

Getting Started

EvenUCan runs entirely in your browser as a Progressive Web App — no install needed. Open it, describe what you want to build, and the AI takes care of the rest.

System Requirements & Installation

Web Version (Recommended)

Open **EvenUCan.html** in any modern browser. No install required. Works offline after first load. Chrome / Edge / Firefox all supported.

Desktop App

Download the Windows, macOS or Linux installer from the GitHub Releases page. Installs as a native application with the same features.

Claude API Key

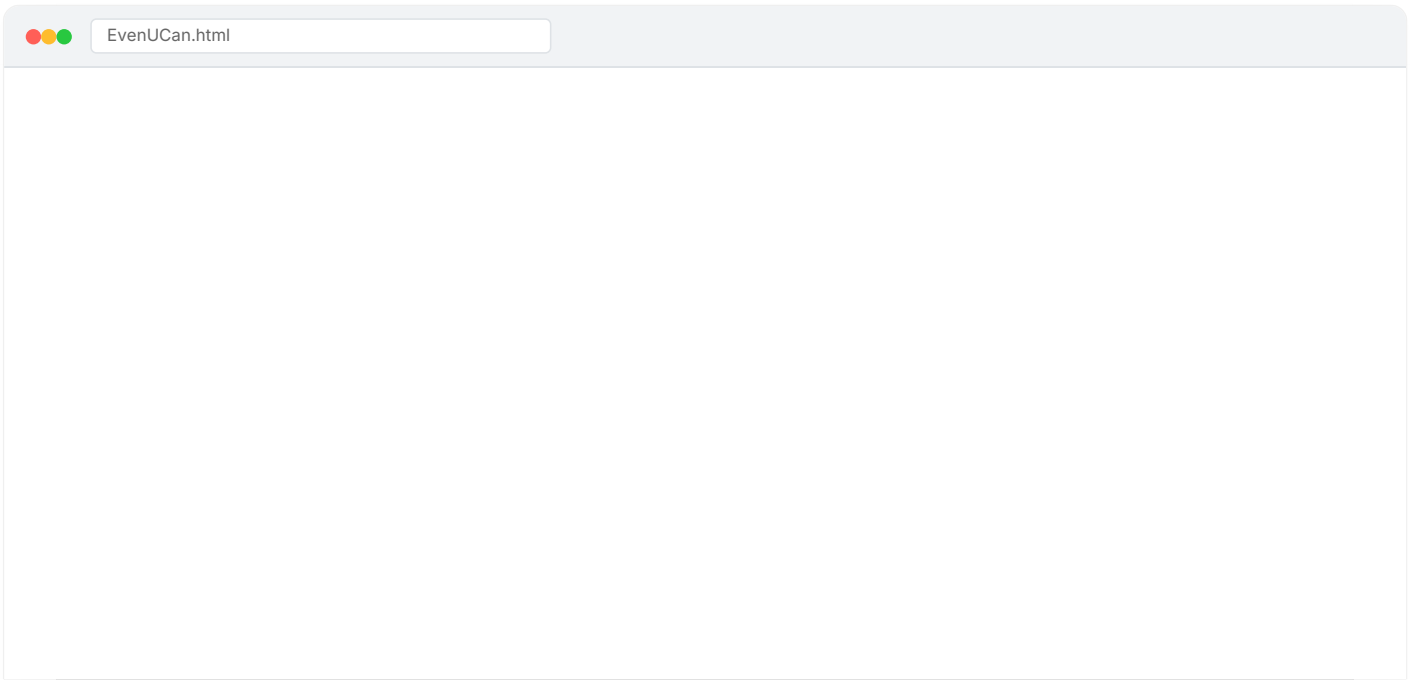
AI features require a free Claude API key from **console.anthropic.com**. The key is stored only in your browser — never sent to any server other than Anthropic.

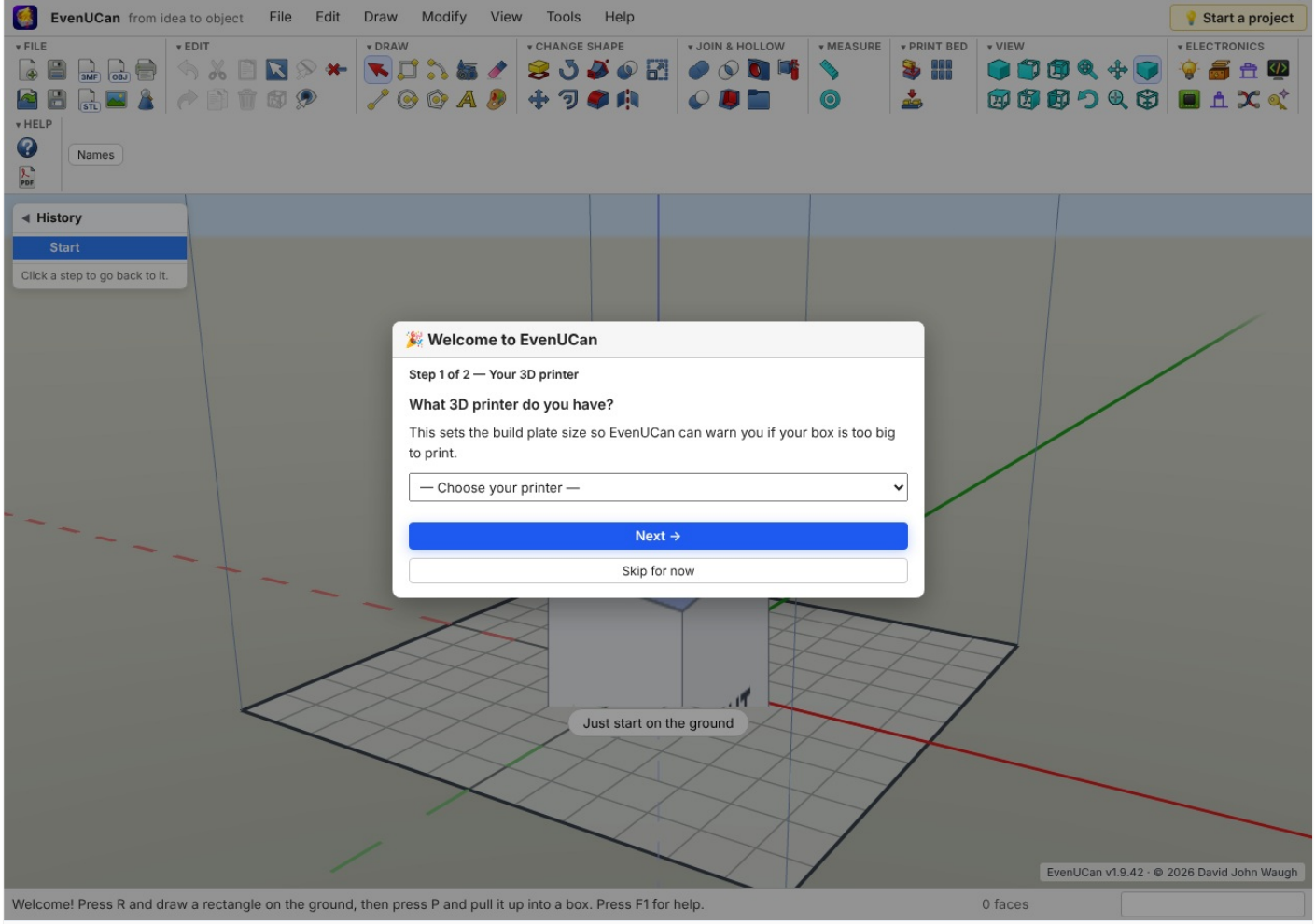
3D Printer

Any FDM printer works. The app supports Prusa, Voron, Bambu, Ender 3 and custom bed sizes. PLA or PETG recommended for enclosures.

First Launch — Printer Setup

When you open EvenUCan for the first time you'll see the **Printer Setup** dialog. This tells the app how big your printer's build plate is so it can automatically fit enclosures to your machine.





First launch: set your printer's build volume, units, and view mode. This is remembered for every future session.

1 Choose your printer

Select from the preset list (Voron 0, Prusa Mini, Ender 3, Prusa MK4, Bambu X1, Voron 2.4) or enter custom X, Y, Z dimensions in millimetres.

2 Set your preferred units

Choose between Millimetres, Centimetres, Metres, Inches, or Feet & inches. This applies globally to all dimensions in the app.

3 Pick a view mode

Perspective (natural) gives a realistic 3D view. **Parallel** is better for precise measurements and technical drawings.

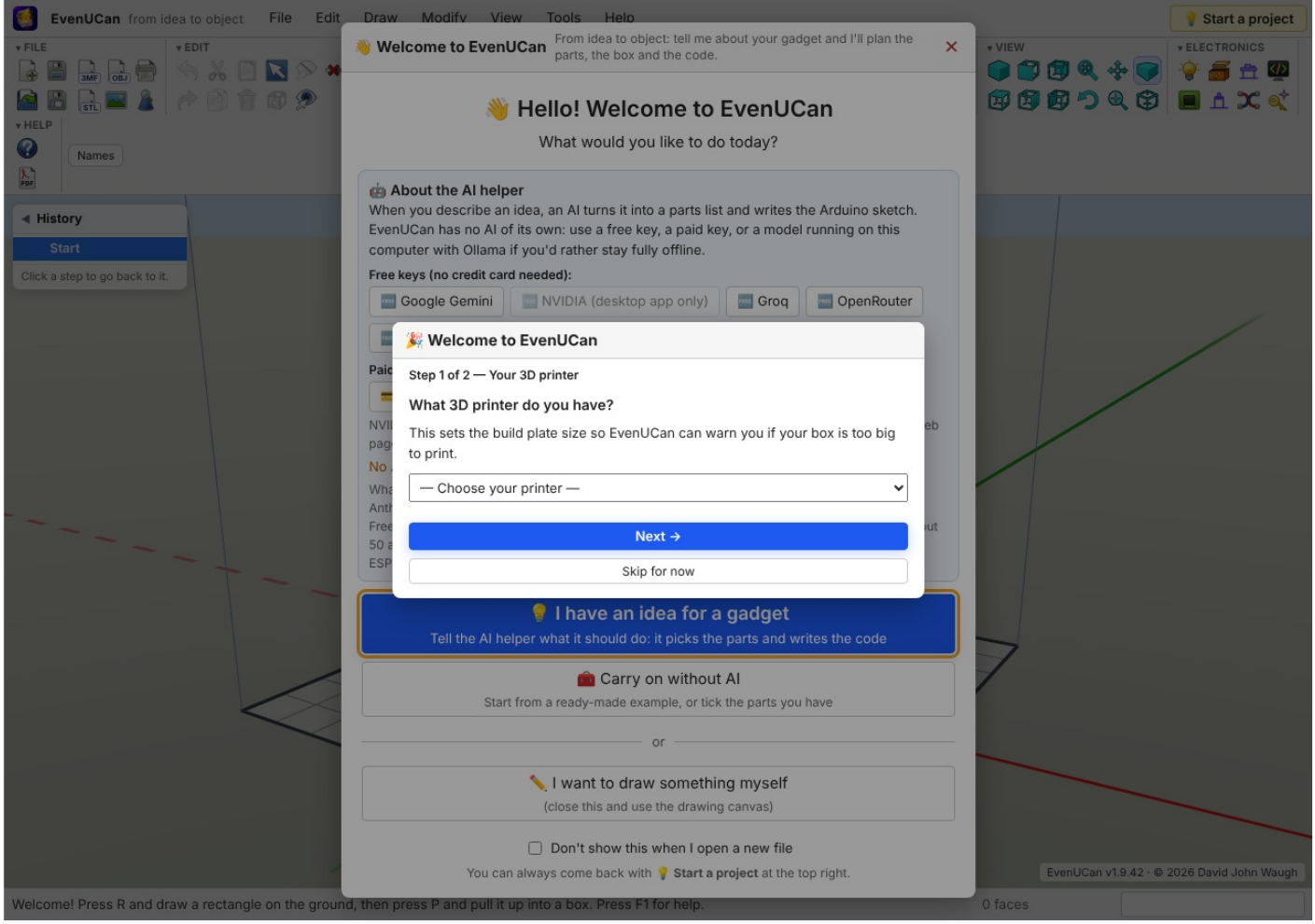
4 Click OK

Your settings are saved in the browser and remembered for every future session. You can change them later via **File → Printer size**.

The Interface

EvenUCan has three main areas:

EvenUCan.html



The main interface: menu bar across the top, 3D canvas in the centre, and tool panel on the left.

Top Menu Bar

- **File** — New, Open, Save, Export (3MF, STL, OBJ, PNG)
- **Edit** — Undo, Redo, Select All, Copy, Paste
- **Draw** — Add shapes: Box, Cylinder, Sphere, Text
- **Modify** — Transform, Boolean ops (Union, Subtract)
- **View** — Reset camera, Zoom Fit, Grid on/off, Wireframe
- **Tools** — Box Designer, Parts Library, Export Settings
- **Help** — Documentation, keyboard shortcuts, About

Left Tool Panel

The left panel changes depending on the active tool:

- **Start a project** — AI wizard for new projects
- **File actions** — Quick access to FILE, SAVE, EXPORT
- **Draw tools** — DRAW, CHANGE SHAPE, JOIN & HOLLOW
- **Measure** — Ruler tool for checking dimensions
- **Electronics (ELECTRO)** — Parts library and PCB placement

> **Quick start tip**
> — The quickest way to explore the app is to click `Start a project` in the top bar. This opens the AI wizard that guides you step-by-step from an idea to a finished enclosure design.

CHAPTER 2

Starting a Project with AI

The AI wizard is the heart of EvenUCan. Describe your electronics project in plain language and Claude AI generates a complete parts list, wiring diagram, and 3D enclosure design.

Your Claude API Key

The AI wizard uses **Claude by Anthropic** to understand your project description and generate designs. You'll need a free API key the first time you use it.

1. **Get your free API key**
   Visit **console.anthropic.com**, create a free account, and generate an API key under *Settings → API Keys*. Free tier includes enough credits for many designs.

2. **Paste it into EvenUCan**
   Click `Start a project`, then paste your key in the API key field. The key is saved in your browser's local storage and never transmitted to anyone except Anthropic's own servers.

> **Your key stays private**
> — EvenUCan is a local web app. Your API key is stored only in your browser and is sent directly to Anthropic when you use AI features. The app has no server of its own and never sees your key.

Describing Your Project

The more detail you give, the better the AI's output. Good descriptions include:

- The **microcontroller** you're using (Arduino Nano, ESP32, Raspberry Pi Pico, etc.)
- The **sensors and modules** you need (DHT22, BME280, OLED, relay, etc.)
- What the device **does** — this helps the AI suggest appropriate additions
- Any **constraints** — battery powered, needs to be waterproof, fits in a pocket, etc.

Example Project Descriptions

Simple: *"ESP32 with DHT22 temperature sensor, OLED display, and a push button to cycle through readings."*

Detailed: *"Battery-powered soil moisture monitor with an Arduino Nano 33 IoT, two capacitive soil sensors, a small OLED screen showing percentage, a green/red LED status indicator, and a buzzer that sounds when soil is too dry. Needs to fit on a window sill."*

Vague (still works): *"A smart plant watering system."* — The AI will make sensible assumptions and you can refine the results.

Reviewing AI Results

After generating, the wizard shows you:

Parts List
Every component with its type, dimensions, mounting position (floor, wall, or outside), and connection pins. Each part can be edited or replaced.

PCB Layout

Components placed on the PCB with their physical positions relative to each other. Drag to rearrange, click to edit any part's properties.

Wiring Diagram

Colour-coded wiring connections between all components. Shows VCC, GND, and signal pins. Can be exported as an image or used as a reference.

3D Enclosure Preview

A custom box sized to fit your components, with mounting holes, cutouts for connectors, and wall-mounted features all pre-positioned.

AI results are a starting point

— Always review the generated parts list carefully. The AI is excellent at suggesting appropriate components but may not know your exact variant (e.g. the specific OLED driver chip). Edit any part that doesn't match your hardware before exporting.

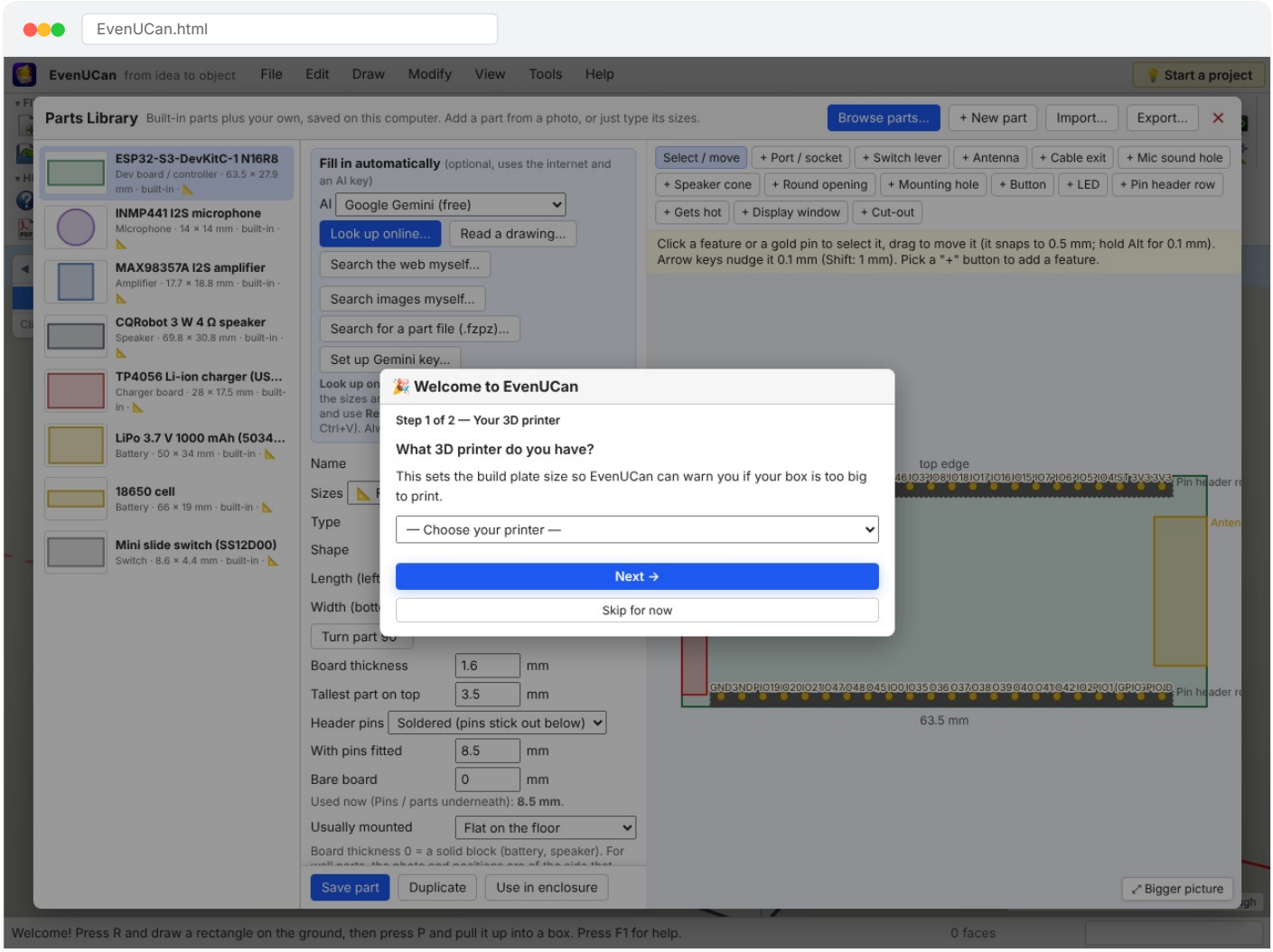
CHAPTER 3

Parts & Electronics

EvenUCan knows the physical dimensions of hundreds of common modules and sensors. The parts library powers both the PCB layout and the 3D box design.

Parts Library

Click the **ELECTRO → Parts** section in the left panel to browse, search, and add components manually. Each part has:



The ELECTRO / Parts panel shows the components in your project. Click any part to edit its properties or position.

Field	Description
Name	Descriptive name, used by the AI and in the parts list
Type	sensor, display, mcu, relay, other — affects rendering style
Dimensions (W × D)	Physical board size in millimetres
Mount	`floor` (inside base), `wall` (on a vertical face), `outside` (externally mounted, e.g. soil probes)
Features	Cutouts and holes in the box: `opening` (round hole), `hole` (standoff mounting hole)
Pins	Connection points used in the wiring diagram

Commonly Matched Modules

EvenUCan automatically recognises these modules when you describe them in plain English:

- SSD1306 / SH1106 OLED displays (various sizes)

- BME280 / BMP280 temperature & pressure sensors
- PIR motion sensors (HC-SR501 and compatible)
- DHT11 / DHT22 / AM2302 humidity & temperature sensors
- 5mm LEDs with resistor FIXED 1.9.42
- WS2812 / NeoPixel addressable LED strips
- Capacitive soil moisture sensors FIXED 1.9.42
- BH1750 light sensors
- Tactile push buttons
- Passive buzzers / piezo speakers
- MPU-6050 / GY-521 accelerometers
- Relay modules (single and multi-channel)

🛠 **Missing a part?**
— Click
Add part manually
in the ELECTRO panel, or use the *Make it with AI* button to auto-generate a YAML definition from a description. The YAML is then saved to your parts library for future use.

PCB & Component Placement

The PCB view shows a top-down view of all components laid out on a virtual board. Components are positioned automatically but can be dragged to any position. The placement here directly affects the 3D box generated in the next step.

Mount Types

- **Floor mount** — Component sits inside the base of the box, flat on the bottom. Used for MCUs, breakout boards, batteries.
- **Wall mount** — Component is mounted on the inside of a vertical wall, facing outward through a cutout. Used for buttons, LEDs, connectors, displays.
- **Outside mount** — Component is physically external to the enclosure. Used for soil moisture probes, antennas, external sensors.

Wiring Diagram

The wiring view draws colour-coded wire connections between all components, based on their pin definitions. VCC connections are red, GND is black, and signal wires use distinct colours per connection.

💡 **Export as image**
— Use
File → Export PNG
while the wiring view is active to save a wiring diagram you can print and keep next to your build.

CHAPTER 4

3D Box Design

EvenUCan generates a custom enclosure sized exactly for your components, complete with mounting holes, wall cutouts, and a snap-fit lid.

Auto-Generated Enclosure

After the AI generates your parts list, click **Generate base + lid** to create a custom 3D enclosure. The box is sized to comfortably fit all your floor-mounted components with appropriate margins, and wall cutouts are automatically added for all wall-mounted parts.

Base

The main box body with floor standoffs for mounting the PCB, and wall cutouts for buttons, LEDs, connectors, and displays.

Lid

A snap-fit lid designed to match the base. Can be friction-fit or use corner posts depending on your design.

Wall Cutouts

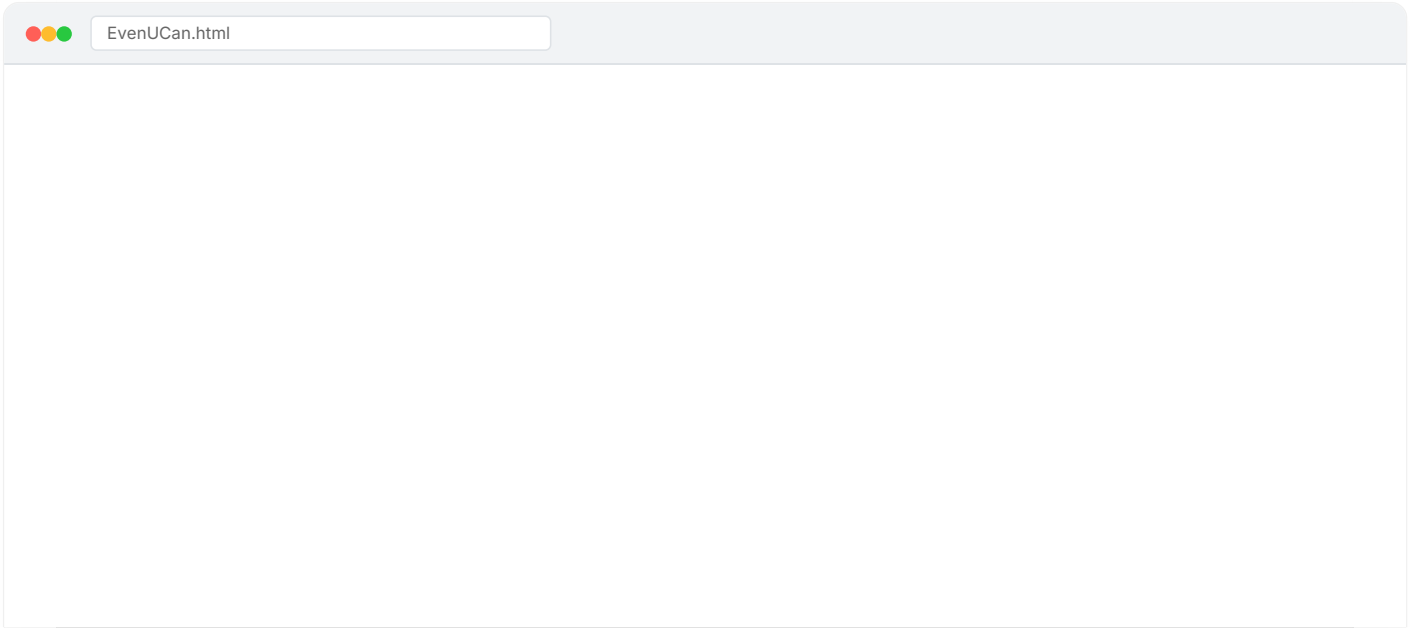
Round holes are automatically sized and positioned for LEDs and round openings. Rectangular cutouts are added for displays, connectors, and buttons.

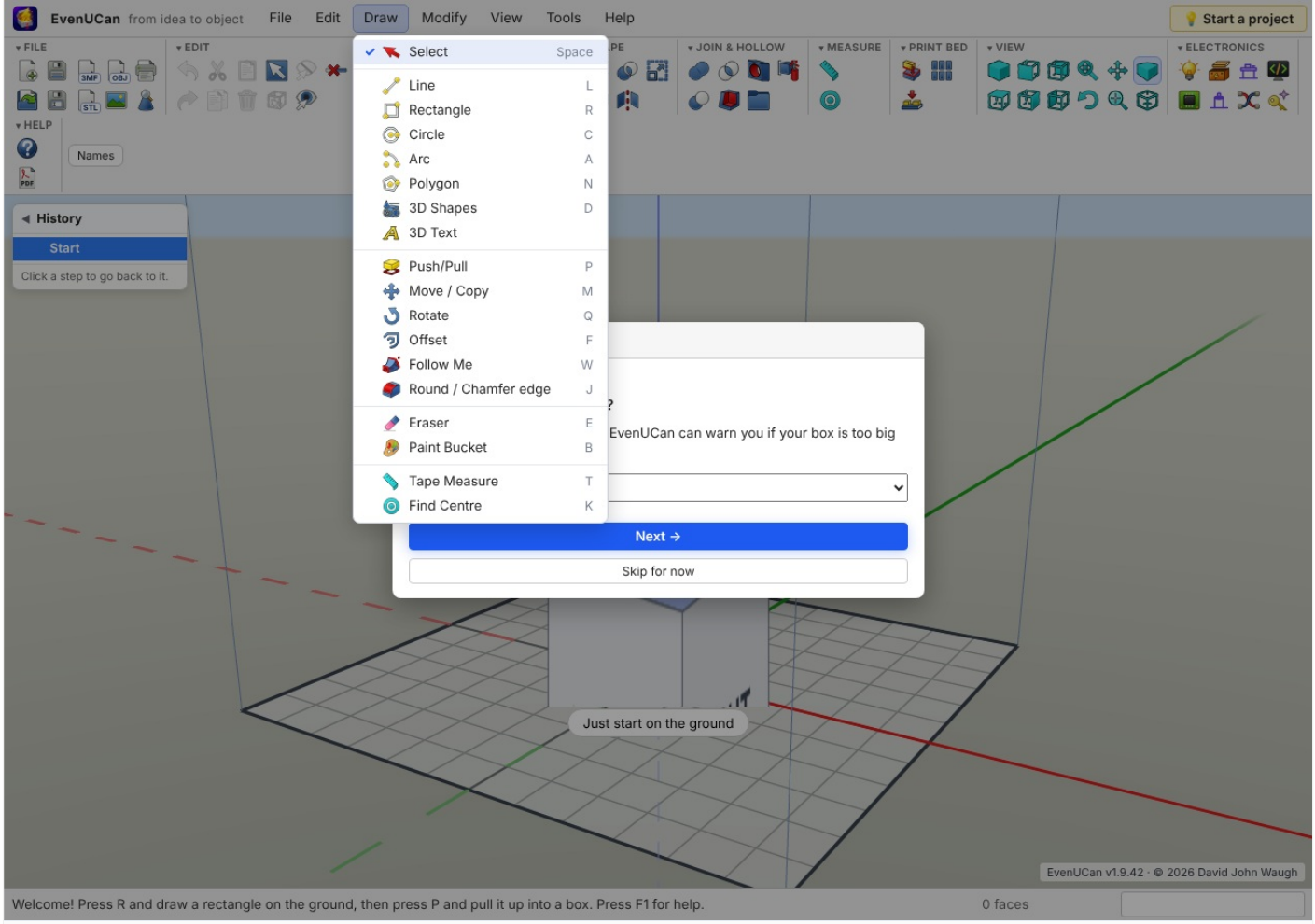
Standoffs

M2 or M3 standoffs (or heat-set insert mounts) are added for the MCU and other floor-mounted boards that need secure attachment.

Manual Drawing Tools

You can also draw and modify the enclosure manually using the **Draw** and **Modify** menus:





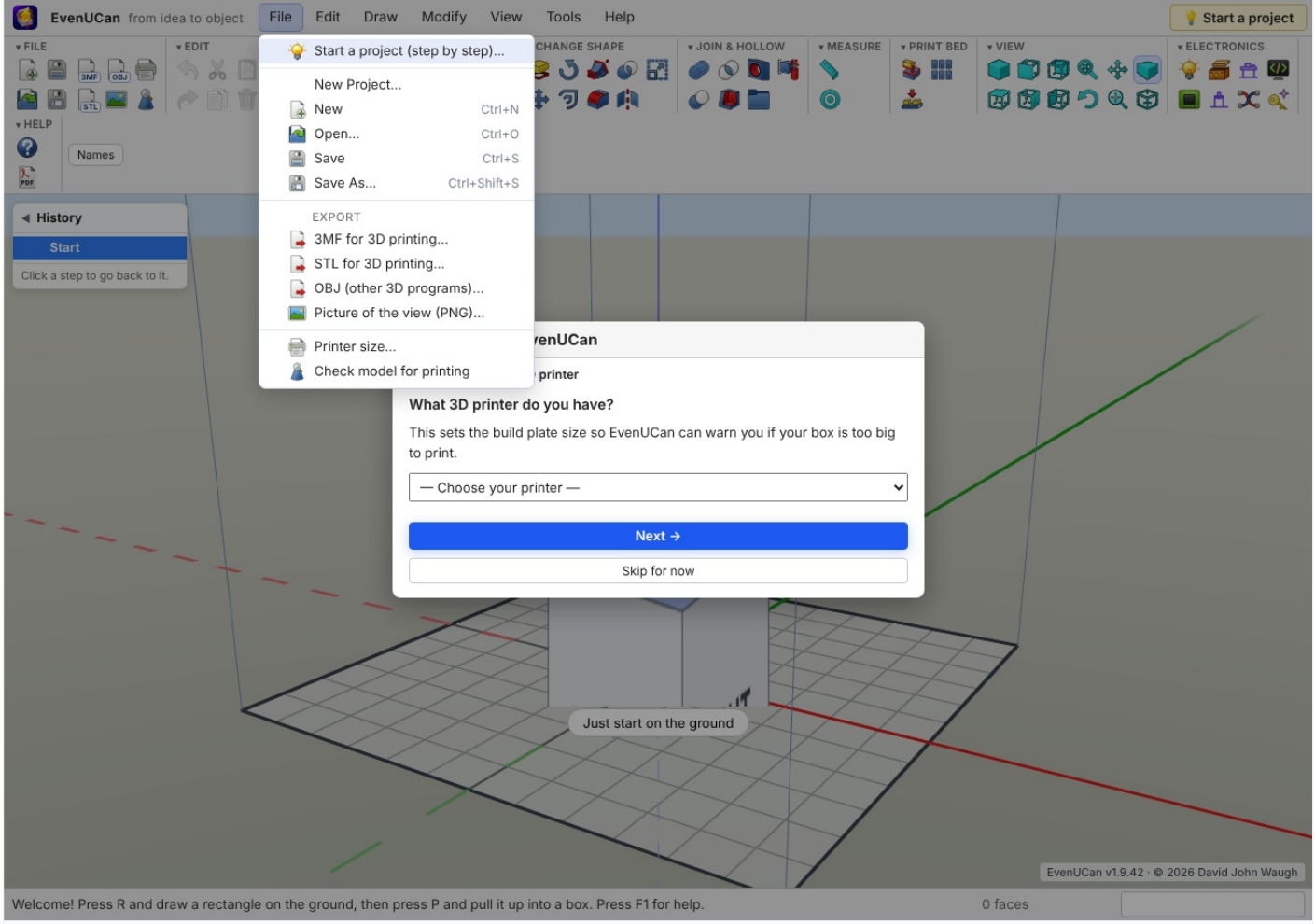
The Draw menu gives access to primitive shapes, 3D text, and special electronic features (holes, standoffs, board mounts).

Draw Tool	Purpose
Box / Rectangle	Draw a rectangular solid or the outer shell of your enclosure
Cylinder / Circle	Add cylindrical features, connectors, or round cutouts
Hole	Subtract a hole through any surface — for screws, LEDs, buttons, cables
Stand-off for heat-set insert	PCB mounting post designed for M2/M3 heat-set inserts
Board mount	Quick mounting feature for standard board sizes
3D text	Emboss or deboss labels, logos, or descriptions into any surface

Exporting for 3D Printing

When you're happy with the design, export it for your slicer:

EvenUCan.html



Export options are in the File menu: 3MF (recommended), STL, OBJ for other 3D tools, or PNG screenshot of the current view.

1 File → 3MF for 3D printing (recommended)

3MF is the modern 3D printing format and is better than STL for most slicers. Supported by PrusaSlicer, Bambu Studio, Cura, and Orca Slicer.

2 Open in your slicer

Import the 3MF file into your slicer. The base and lid will appear as separate objects so you can slice them together or separately.

3 Slice and print

Use standard settings: 2–3 perimeters, 20–25% infill, no supports needed (the design avoids overhangs). PLA works well; PETG if you need more heat resistance.

> 💡 **STL is also available**
>
> via *File → STL for 3D printing* for older software. Use OBJ if you need to open the model in Blender, Fusion 360, or another 3D modelling tool.

CHAPTER 5

YAML / Code Export

Every component in your project can be exported as a YAML definition — a portable file that describes the part's physical dimensions, mounting, and pins.

The YAML export is useful for:

- Saving a custom part definition for use in future projects
- Sharing parts with the community
- Importing into the parts library
- Version-controlling your electronics project

EXAMPLE YAML — CAPACITIVE SOIL MOISTURE SENSOR

```yaml
id: p-cm-soilmoist
name: Capacitive soil moisture sensor module
type: sensor
dimensions:
  width: 22
  depth: 35
  pcb: 1.6
  height: 3
  below: 3
mount: outside        # Mounts externally (probe goes outside the box)
pins:
  - name: VCC
    rail: 3v3
  - name: GND
    rail: gnd
  - name: AOUT
    rail: sig
features: []          # No wall cutout needed (mounts outside)
```

Making Parts with AI

Can't find a part in the library? Use the **Make it with AI** button in the parts panel to describe any component in plain English. The AI generates a complete YAML definition including dimensions, mounting type, and pin connections.

1. **Click "Make it with AI"**
   In the parts panel, click the button and describe your component: *"DS18B20 waterproof temperature probe on a 3-wire cable"*

2. **Review the YAML**
   The AI generates a YAML definition. Check the dimensions match your actual component — you may need to adjust width and depth.

3. **Click "Make it with AI" again to apply**
   A second click sends the YAML to the box designer, adding the new part to your project. The box will automatically resize and add any needed cutouts.

CHAPTER 6

Tips & Keyboard Shortcuts

Work faster with these keyboard shortcuts and hidden features.

Navigation & View

Shortcut	Action
`Mouse drag`	Rotate the 3D view
`Scroll wheel`	Zoom in / out
`Right-click drag`	Pan the view
`F`	Zoom to fit all objects
`Iso` button	Reset to isometric view
`Top` / `Front` / `Side`	Snap to orthographic views

Editing

Shortcut	Action
`Ctrl` + `Z`	Undo
`Ctrl` + `Shift` + `Z`	Redo
`Ctrl` + `S`	Save project
`Ctrl` + `N`	New project
`Ctrl` + `O`	Open project
`Delete`	Delete selected object
`Escape`	Deselect / cancel current tool
`Ctrl` + `A`	Select all

Pro Tips

Measure before you print

Use the Measure tool (MEASURE section in the left panel) to verify critical dimensions before exporting. Check that the box fits your printer's build volume.

Add tolerance to wall openings

3D-printed holes are usually 0.2–0.4mm smaller than the design. EvenUCan adds a small tolerance automatically, but if you're printing at a lower quality, manually increase hole diameters by 0.3mm.

Save frequently

Use `Ctrl` + `S` to save your project as a JSON file. This saves the full state including parts, positions, and AI-generated data — not just the 3D shape.

Soil sensors and external components

Parts marked `mount: outside` (like soil moisture probes) appear in the parts list but don't take up space inside the box. The box is only sized for interior components.

CHAPTER 7

What's New in 1.9.42

Version 1.9.42 brings three important fixes to the parts matching system, improving accuracy for LED and soil sensor detection.

🔧 Smarter Parts Matching

The AI-assisted parts detection engine was updated with three fixes that improve how common components are identified from natural language descriptions.

Bug Fixes

1. **LEDs no longer misidentified as soil moisture sensors**

   When an LED was described as a "soil-dry indicator LED" or similar, the word "soil" in the description could cause it to be matched as a capacitive soil moisture sensor. Fixed by reordering the matching priority: LED patterns are now checked before soil moisture patterns, and the soil moisture matcher explicitly excludes terms like "LED" and "indicator".

2. **LED wall mounts now have a physical hole**

   LEDs placed on a wall were rendering a visual outline on the box face but not cutting through the wall — meaning the actual LED could not shine through. Fixed by changing the LED feature type from `t:'led'` (decorative only) to `t:'opening'` (physical cutout, same type used by PIR sensor domes).

3. **Soil moisture sensors now mount outside the box**

   Capacitive soil moisture probes were being placed inside the enclosure with `mount:'floor'`. Since the sensing probe must extend into the soil, the sensor is now marked `mount:'outside'` — it appears in the wiring diagram but does not take up interior space in the box, and no wall cutout is added.

> ℹ **Existing projects**
> — If you have saved projects from v1.9.41 that include LEDs or soil sensors, regenerate the parts list (or manually edit the mount type) to pick up these fixes.

Previous Recent Changes

Version	Highlights
1.9.41	Added WS2812/NeoPixel LED strip support; improved BME280 vs BMP280 detection
1.9.40	MPU-6050 accelerometer support; relay module detection
1.9.39	BH1750 light sensor; improved buzzer/piezo matching
1.9.38	Multi-page wizard; improved AI response streaming

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Source code: **github.com/bullbrand/EvenUCan** · Released under MIT Licence

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